Geon Choi

Ph.D. Student, Kim Jaechul Graduate School of AI, KAIST

choigeon@kaist.ac.kr | Homepage | GitHub | LinkedIn

Research Interests

- **Multimodal AI:** Multimodal Large Language Models (MLLMs) and Vision-Language-Action (VLA) Models
- **AI for Healthcare:** Medical Vision-Language Models, Medical Imaging

Education

Korea Advanced Institute of Science and Technology (KAIST) — Feb. 2026 – Present
Ph.D. in Artificial Intelligence (Advisor: Prof. Edward Choi) — Seongnam, South Korea

Korea Advanced Institute of Science and Technology (KAIST) — Feb. 2024 – Feb. 2026
M.S. in Artificial Intelligence (Advisor: Prof. Edward Choi) — Seongnam, South Korea

Pohang University of Science and Technology (POSTECH) — Feb. 2020 – Feb. 2024
B.S. in Electrical Engineering — Pohang, South Korea

- GPA: 4.04/4.3 (Summa Cum Laude); ranked 2nd in the Department of Electrical Engineering

Jeonbuk Science High School — Mar. 2018 – Feb. 2020
Early Graduation — Iksan, South Korea

- Ranked 3rd; awarded the POSTECH Presidential Fellowship

Publications

C: Conference, P: Preprint, *: Equal Contribution

[C3] **Language: A Benchmark for Structured and Sequential Chest X-ray Interpretation**
Jong Hak Moon, **Geon Choi**, Paloma Rabaey, Min Gwan Kim, Hyuk Gi Hong, Jung-Oh Lee, Hangyul Yoon, Eun Woo Doe, Jiyoun Kim, Harshita Sharma, Daniel C. Castro, Javier Alvarez-Valle, Edward Choi
Conference on Health, Inference, and Learning (CHIL), 2026 [arXiv] [Code]

[C2] **Instruction-Guided Lesion Segmentation for Chest X-rays with Automatically Generated Large-Scale Dataset**
Geon Choi*, Hangyul Yoon*, Hyunju Shin, Hyunki Park, Sang Hoon Seo, Eunho Yang, Edward Choi
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 [arXiv] [Code]

[C1] **CXReasonBench: A Benchmark for Evaluating Structured Diagnostic Reasoning in Chest X-rays**
Hyungyung Lee, **Geon Choi**, Jung-Oh Lee, Hangyul Yoon, Hyuk Gi Hong, Edward Choi
Conference on Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track, 2025 **Spotlight**
[arXiv] [Code]

[P2] **XFlow: A Workflow Model for Instruction-Guided Lesion Segmentation in Chest X-rays**
Geon Choi, Hangyul Yoon, Hyunki Park, Sang Hoon Seo, Edward Choi
arXiv preprint, 2026 [arXiv]

[P1] **CheXpercept: A Benchmark for Evaluating Expert-Level Lesion Perception in Chest X-rays**
Geon Choi, Hangyul Yoon, Nalee Kim, Jeong Yun Jang, Hyunju Shin, Hyunki Park, Sang Hoon Seo, Edward Choi
arXiv preprint, 2026 [arXiv]

Research Experience

KAIST AI Research Internship (KAIRI) — Jun. 2023 – Aug. 2023
BioImaging, Signal Processing and Learning Lab, KAIST (Advisor: Prof. Jong Chul Ye) — Daejeon, South Korea

- Generative models for solving inverse problems in medical imaging

Undergraduate Research Internship — Jan. 2023 – Jun. 2023
Bio Optics and Acoustics Lab, POSTECH (Advisor: Prof. Chulhong Kim) — Pohang, South Korea

- Computer-aided diagnosis system for breast ultrasonography

Honors and Awards

KAIST Scholarship, KAIST 2024 – Present
Tuition scholarship

KEC Scholarship, KEC Foundation 2023
Merit scholarship for outstanding EE students ($2k)

National Science & Technology Scholarship, KOSAF 2022 – 2023
Full tuition scholarship

POSTECH Jigok Scholarship, POSTECH 2020 – 2021
Full tuition scholarship

POSTECH Presidential Fellowship, POSTECH 2020 – 2023
Merit fellowship for exceptional freshmen ($10k)

Academic Service

Conference Reviewer

- International Conference on Learning Representations (ICLR), 2027

Teaching

Instructor, KoSAIM Summer School Hands-on Session 2024 – 2025
Taught chest X-ray classification with CNNs to medical AI researchers and medical students in South Korea

Course Assistant, Digital System Design (EECE273), POSTECH 2022
Student mentor

Skills

- **Programming Languages:** Python, C, MATLAB
- **Core Frameworks & Libraries:** PyTorch, Transformers, scikit-learn
- **MLLM Training & Serving:** TRL, verl, vLLM, DeepSpeed
- **Embodied AI:** starVLA
- **Tools & Infrastructure:** Git, Linux, TensorBoard, Weights & Biases
- **Languages:** Korean (native), English (professional working proficiency)

References

Edward Choi, Associate Professor, Kim Jaechul Graduate School of AI, KAIST
M.S. and Ph.D. Advisor | edwardchoi@kaist.ac.kr